

6

Broadband Wire Arrays

6.1 INTRODUCTION

It was pointed out in Chapter 2 that arrays possess unique features from a radar cross section point of view. To reduce the RCS outside the operating band of an antenna in general, a bandpass radome is often placed in front of it (see Chapter 2, Fig. 2.1). In the case of an array we recall from Chapter 2 that a low RCS is obtained when the terminal reflection coefficient is low; in other words, the bandwidth of the array should ideally exceed that of the radome. Furthermore, due to the high price tag of arrays in general, it is desirable to pass on as much information through them as possible. Thus, we shall in this chapter consider the principles for broadband arrays, which is of interest to the communication community.

Quite often arrays are expected to be rather narrowbanded. This is usually rooted in the fact that a single dipole has a rather limited bandwidth (10–20% unless broadband-compensated; see Appendix B). However, when placed in an array the mutual coupling between the individual dipoles can as shown here be designed to increase the bandwidth substantially rather than reduce it (see also Appendix D). Furthermore, addition of a groundplane in the back and dielectric compensation slabs in the front can lead to designs where the bandwidth is approaching a decade rather than an octave. However, it should be emphasized that merely using a couple of dielectric slabs laying around in the shop will not do. Proper design is of utmost importance.

It should also be pointed out at the beginning of this chapter that frequent use will be made of rather simple equivalent circuits. These are intended only

to explain the physics of the problem and guide us in our design effort. At no time do we rely on these to obtain actual calculated values. These are obtained rigorously by use of either the PMM (infinite $\times$ infinite array) or the SPLAT program (finite $\times$ infinite array).

Everybody can learn to run somebody else's program. Only a minority is capable of making the correct interpretation. But a simple equivalent circuit certainly helps.

6.2 THE EQUIVALENT CIRCUIT

The key to the equivalent circuit for an array of dipoles backed by a groundplane and with one or more dielectric slabs in the front is obtained from Problem 5.2 in reference 92. For easy reference it has been reproduced in Fig. 6.1. The circuit is strictly speaking only valid for the E and H planes, and for no grating lobes. Furthermore, the elements must be short as stated in the original problem [92]. In the intercardinal planes some cross-polarization will occur as is in general the case with dipole arrays. However, that should not deter us from developing our designs guided by this equivalent circuit since we ultimately would use the PMM code for verification of the concept for all scan planes.

Furthermore, R_{A0} denotes the radiation resistance of the array when located in an infinite free space and with no groundplane. Similarly, R_{A1} denotes the radiation resistance of the same array without a groundplane and located in an infinite medium with intrinsic impedance Z_1 .

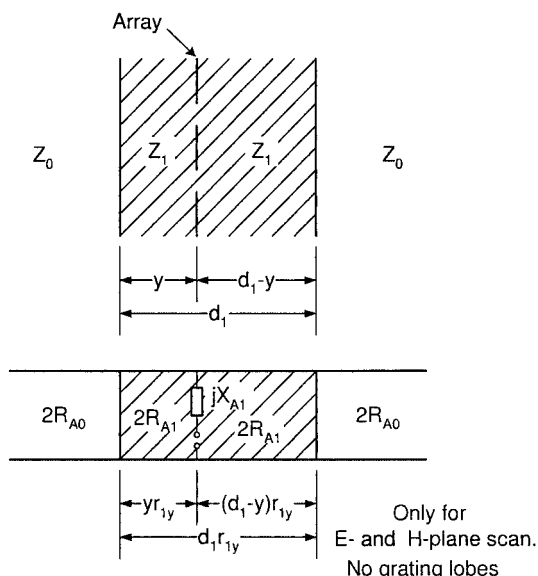


Fig. 6.1 This problem is taken from reference 92 and constitutes the entire basis for this chapter about broadband wire arrays. The elements are assumed to be short in terms of wavelength.

We see no reason to reveal the solution to Problem 5.2 (of reference 92) and thereby deprive the reader of a great opportunity to enlighten himself.

6.3 AN ARRAY WITH GROUNDPLANE AND NO DIELECTRIC

To start, let us consider an array of dipoles with a groundplane and no dielectric slabs in front. From Fig. 6.1 we readily obtain the equivalent circuit for this case as shown in the insert of Fig. 6.2. We shall point out various impedance

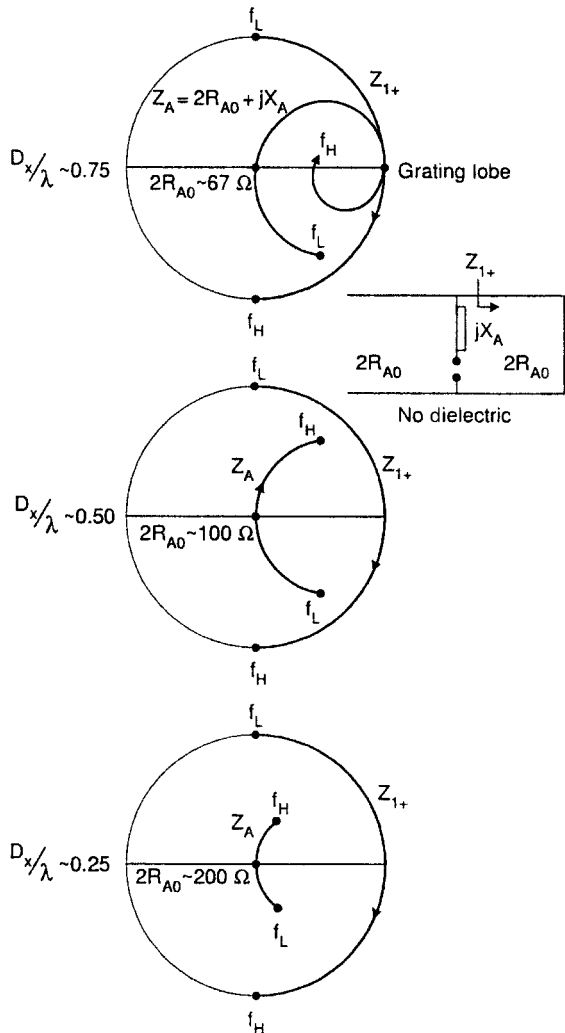


Fig. 6.2 Typical terminal impedance $Z_A = 2R_{A0} + jX_A$ in the negative direction. The interelement spacing is varied from $D_x/\lambda = 0.75$ to 0.25. The groundplane impedance Z_{1+} is purely imaginary; that is, it is located on the rim of the Smith charts as shown.

components. First, the impedance Z_{1+} denotes the input impedance of the equivalent transmission line looking in the positive direction from the array and terminated in a short (namely the groundplane). This impedance will always be purely imaginary; that is, it will be located on the rim of the Smith chart as also shown for various cases in Fig. 6.2 to be discussed later. Quite often it is suggested that this impedance ideally should be infinite (see *Common Misconceptions*, Section 6.12). This will happen when the spacing between the elements and the groundplane is one-quarter wavelength (for broadside radiation). At a lower frequency f_L , it becomes inductive and similarly it is capacitive at the higher frequency f_H . Looking in the negative direction we observe an infinite transmission line with characteristic impedance $2R_{A0}$. Finally, the reactive part jX_A of the antenna impedance $Z_A = 2R_{A0} + jX_A$ in the negative direction is obtained only from the evanescent waves surrounding the elements on both the positive and negative sides of the array elements, see reference 93.

Let us now examine the typical effect of the interelement spacing D_x as illustrated for three cases in Fig. 6.2. At the top we show a typical impedance Z_A for $D_x/\lambda \sim 0.75$. This is followed by cases for $D_x/\lambda \sim 0.50$ and 0.25 as shown in the middle and the bottom, respectively. Note that $2R_{A0}$ is approximately inverse proportional to D_x/λ and also that more bandwidth is obtained for smaller interelement spacings. The latter observations are counterintuitive to many people since the mutual impedance certainly will increase with smaller interelement spacings. (Note that the real part of the impedance does indeed increase as D_x decreases while the imaginary part does not.)

One physical explanation for this fact is that the imaginary part of the mutual impedances start canceling each other as we move away from the reference element, provided that the interelement spacings are small ($<0.4\lambda$), while they add up (eventually to disastrous grating lobes) if D_x/λ is larger than ~ 0.5 . For further discussion of this subject see Section 6.12 and also Appendix D.

6.4 PRACTICAL LAYOUTS OF CLOSELY SPACED DIPOLE ARRAYS

When laying out the elements of a broadband dipole array without a groundplane, one should keep in mind that the ideal aperture is really a continuous current sheet as suggested by Wheeler [94]. It has no reactance (when there is no groundplane), only a radiation resistance that is scan-dependent, but can be compensated to a degree with dielectric slabs [95–97]. The question then becomes how best to make an approximation to such a current sheet. One approach is shown in Fig. 6.3. It basically consists of long closely spaced wires fed periodically along the wires as shown. This approach will ensure a current distribution between the feedpoints that is sinusoidal, but with a strong DC component. Thus, the current distribution is significantly different from that of isolated short dipoles that essentially go to zero at the ends (with a somewhat blunt “nose” caused by the end capacity). However, since short wires are associated with inductances, we have inserted series capacitors with the same periodicity as the feed

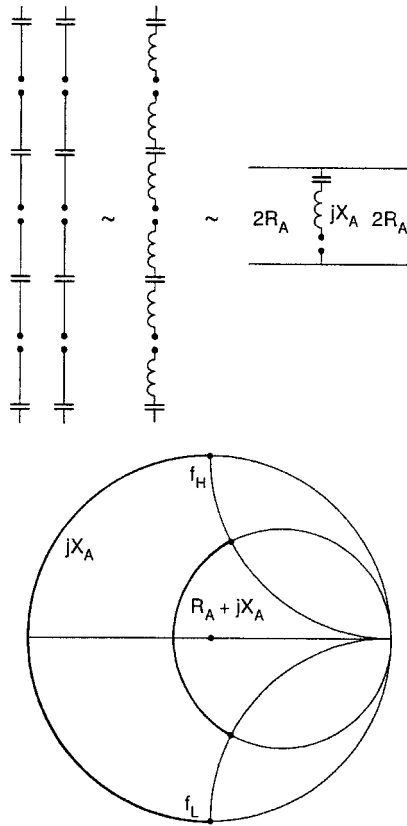


Fig. 6.3 One possible implementation of a current sheet aperture by using infinitely long wires periodically fed and with their inductance compensated by periodically inserting series capacitors. Note: Our goal is to obtain reasonable constant currents along the wires to imitate a current sheet.

points. The resulting equivalent circuit is shown in the insert of Fig. 6.3; and by consulting Appendix B, we readily obtain the locus for the antenna impedance $Z_A = R_A + jX_A$ as shown in the Smith chart in Fig. 6.3. Note that the value of the capacitors is not critical since they are actually in parallel with the tip capacitance between the elements (that can be more substantial than many expect; see also Section 6.12.2 and the discussion concerning circuit analog absorbers [98] as well as Weber [89]).

An alternate approach is shown in Fig. 6.4, left, where the capacitance between the element tips has been increased by tilting the elements as shown. This layout was first presented in Tom Kornbau's dissertation [99] and resulted in an FSS with extreme bandwidth, commonly referred to as the "gangbuster surface" [91]. It was the result of a brainstorm that Tom and the author had one day. However, somebody missed the deadline for the patent applications (some lawyers are fast, others are not).

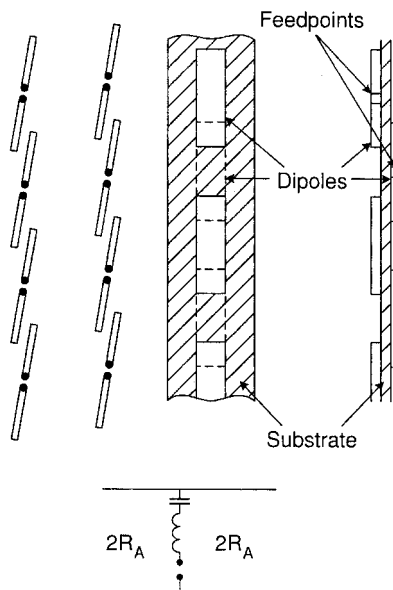


Fig. 6.4 Alternative approaches for designing a current sheet aperture. Left: Tilted and closely packed dipoles. Right: Dipoles alternating between the two sides of a dielectric substrate.

Still another variation is shown in Fig. 6.4, right. There the elements are placed on both sides of a thin dielectric substrate in one interlacing pattern, as shown. That approach is currently being used by Mission Research Corporation, Dayton, Ohio. Harris uses elements loaded with finger capacitors (see Fig. 10.4 and associated comments).

6.5 COMBINATION OF THE IMPEDANCE COMPONENTS

We next show in Fig. 6.5 how the various impedance components are combined. First we connect Z_{1+} in parallel with $2R_{A0}$. Since Z_{1+} is purely reactive, the parallel combination $2R_{A0} \parallel Z_{1+}$ will be located somewhere on a circle going through $2R_{A0}$ and the origin (0,0) of the Smith chart (for a comprehensive introduction to impedance matching in general and broadband matching in particular, see Appendix B). Finally, we add the reactance jX_A in series with $2R_{A0} \parallel Z_{1+}$. Since jX_A is purely imaginary, all the new impedances will lie somewhere on the circles going through the respective impedances $2R_{A0} \parallel Z_{1+}$ and the infinity point in the Smith chart (again consult Appendix B). From Fig. 6.2 we recall that jX_A typically is negative at frequencies below the center frequency f_0 . Thus, the impedances $2R_{A0} \parallel Z_{1+}$ will be dragged downward toward the capacitive region. Similarly, the impedances $2R_{A0} \parallel Z_{1+}$ will be pushed upward when the frequency is above f_0 . The net result is that we obtain a total impedance that is clustered in a smaller area than the impedances of the various impedance components

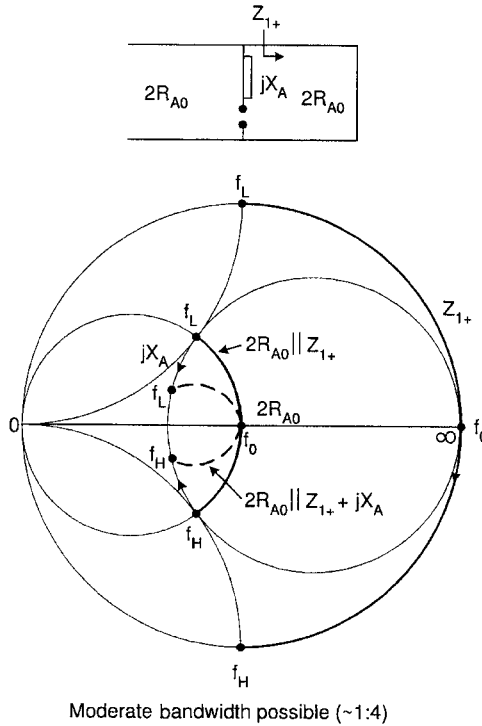


Fig. 6.5 Top: The equivalent circuit for an array of wire dipoles backed by a groundplane. Bottom: The groundplane impedance Z_{1+} at the rim of the Smith chart is being connected in parallel with $2R_{A0}$ to the left and denoted $2R_{A0} \parallel Z_{1+}$. Finally, adding the antenna reactance jX_A in series is seen to produce a more compact terminal impedance $2R_{A0} \parallel Z_{1+} + jX_A$ than without a groundplane.

individually. If we at the same time can land this cluster at the correct impedance, we have indeed obtained a broadband array. Obviously the terminal impedance curve can be compressed further than is shown.

6.6 HOW TO OBTAIN GREATER BANDWIDTH

An array of dipoles with small interelement spacings and backed by a groundplane can be designed to have a bandwidth of around 4:1 with a VSWR < 2 . In order to obtain a greater bandwidth (or a lower VSWR), several ideas come to mind. First, it is desirable that the level of the groundplane impedance Z_{1+} should be high. (This does not mean that it should be infinite at all frequencies; in fact, that is not even desirable in order to compensate other impedance components as shown above. See also Section 6.12 for further discussion.) One approach often suggested is to somehow make a groundplane that would move electrically as a function of frequency with respect to the dipole array. Since that

the groundplane impedance Z_{1+} unaffected, it would be equivalent to making Z_{1+} relatively higher and thereby lead us to a potentially larger bandwidth.

One possible execution of that idea is discussed next.

6.7 ARRAY WITH A GROUNDPLANE AND A SINGLE DIELECTRIC SLAB

In Fig. 6.6 we show an array comprised of a groundplane and a dielectric slab in front of the dipoles. Again, we shall examine the various impedance components and their combinations. First of all, the impedance Z_{1-} denotes the input impedance of the equivalent transmission line at the plane of the array looking in the negative direction. It is seen merely to be the input impedance of a transmission line with characteristic impedance $2R_{A1}$ and terminated in $2R_{A0}$. The Smith chart at the bottom of Fig. 6.6 is normalized to $2R_{A0}$; that is, the transforming circle will for $\beta_1 d_1 = 0$ start from the center of the Smith chart. Furthermore, when the slab is equal to $\lambda_1/4$ at the center frequency f_0 , the input impedance will be determined by the point $(2R_{A1})^2/2R_{A0}$ (for details consult Appendix B). Since $2R_{A1} < 2R_{A0}$ because of the presence of the dielectric, it is clear that the quarterwave point of the transforming circle will be located at a point denoted f_0 to the left of the center of the Smith chart, as shown in Fig. 6.6. Furthermore, the distance between the array and the groundplane is typically chosen also to be equal to one quarter-wavelength at the center frequency f_0 . Thus, $Z_{1+} = \infty$ and will consequently not change Z_{1-} when connected in parallel. Finally, we must according to the equivalent circuit at the top of Fig. 6.6 add jX_A to $Z_{1-} \parallel Z_{1+}$. Typically, $jX_A \sim 0$ at the center frequency f_0 ; that is, the entire terminal impedance $Z_{1-} \parallel Z_{1+} + jX_A$ is essentially given by point f_0 at the center frequency.

Let us next examine the terminal impedance at a typical lower frequency f_L . Again we observe that the impedance Z_{1-} will be located on the same transformation circle as for f_0 above. However, since the electric thickness of the dielectric slab now is smaller than a quarter-wavelength, we will transverse a shorter path than above as indicated by Z_{1-} in the Smith chart in Fig. 6.6. Next, we add the impedance Z_{1+} in parallel with Z_{1-} . Since the frequency f_L is lower than f_0 , Z_{1+} is inductive and will consequently push Z_{1-} up toward the inductive part of the Smith chart along the circle going through Z_{1-} and point (0,0) as indicated by point $Z_{1-} \parallel Z_{1+}$ (for details see Appendix B). Finally, we must add jX_A . At the lower frequency f_L it is capacitive and will consequently push $Z_{1-} \parallel Z_{1+}$ downwards when added in series. It will move along a circle going through $Z_{1-} \parallel Z_{1+}$ and the infinity point to the point denoted f_L (again, see Appendix B for details).

The performance of a typical high frequency f_H is obtained completely analogous to the f_L case. If jX_A is symmetric around f_0 (which is idealized), we will simply find that the terminal impedance is given by the point f_H located symmetrically to f_L . The typical terminal impedance curve as a function of frequency is denoted by $Z_{1-} \parallel Z_{1+} + jX_A$ in Fig. 6.6.

It is interesting to compare the nondielectric case in Fig. 6.5 with the dielectric case shown in Fig. 6.6. A casual inspection indicates that the latter has a greater potential for yielding a wider bandwidth or lower VSWR than the former. The reason is of course that the groundplane impedance Z_{1+} is partly canceled by the reactive part of Z_{1-} at higher as well as lower frequencies. In contrast to the nondielectric case in Fig. 6.5, the Z_{1-} impedance remains equal to $2R_{A0}$ without any reactive part at any frequency and will consequently not reduce the effect of the variation of Z_{1+} as a function of frequency.

The discussion above has been somewhat idealized; for example, $2R_{A0}$ will vary some with frequency and jX_A is not completely symmetric around f_0 .

Input impedance renormalized to $Z_0 = 100 \Omega$ for: dipgp41

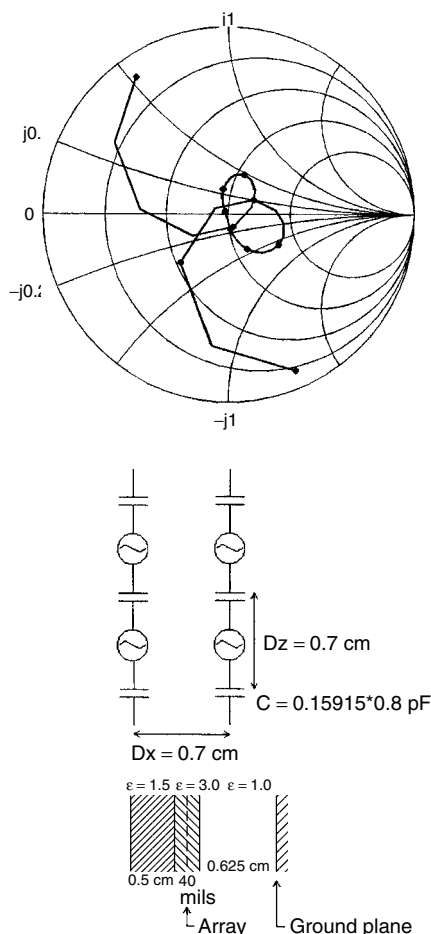


Fig. 6.7 Calculated terminal impedance at broadside obtained from the PMM code based on the dimensions shown in the insert. The values include a small matching section comprised of a transmission line of length 0.13 cm and characteristic impedance 200 ohms.

Nevertheless, it has served our main purpose, namely, to develop a concept yielding a broadband array. The final test will be trying our design using the PMM program. This is the subject of the next section.

6.8 ACTUAL CALCULATED CASE: ARRAY WITH GROUNDPLANE AND SINGLE DIELECTRIC SLAB

In the insert of Fig. 6.7 we show the pertinent dimensions of an array of dipoles with a groundplane and essentially a single dielectric slab (*Note*: The dielectric “underwear” [100] next to the dipole elements serve primarily as protection. It

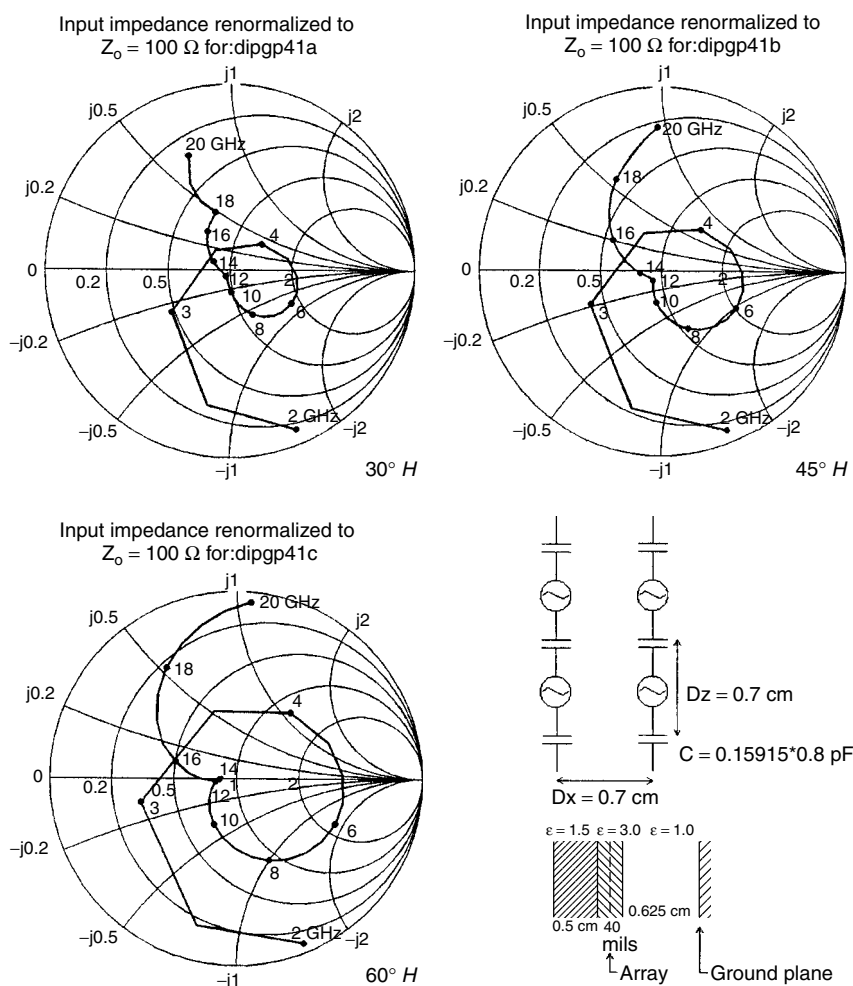


Fig. 6.8 Same array configuration as shown in Fig. 6.7 but for scan angles 30° , 45° , and 60° in the H plane.

is so thin that its transforming contribution is minimal. It is, of course, included exactly in the PMM calculations). The terminal scan impedance as obtained from the PMM program is shown in the Smith chart in Fig. 6.7. Several points should be emphasized when compared to the qualitative case in Fig. 6.6. First the calculated curve in Fig. 6.7 has been normalized to 100 ohms, which centers the impedance curve fairly well. Furthermore, the calculated curve actually is the input impedance of a transmission line with characteristic impedance 200 ohms, length 0.13 cm, and termination in the actual scan impedance. This transmission line serves the sole purpose of improving the matching as explained in Appendix B. Although short (the electric length amounts to 0.009λ at 2 GHz)

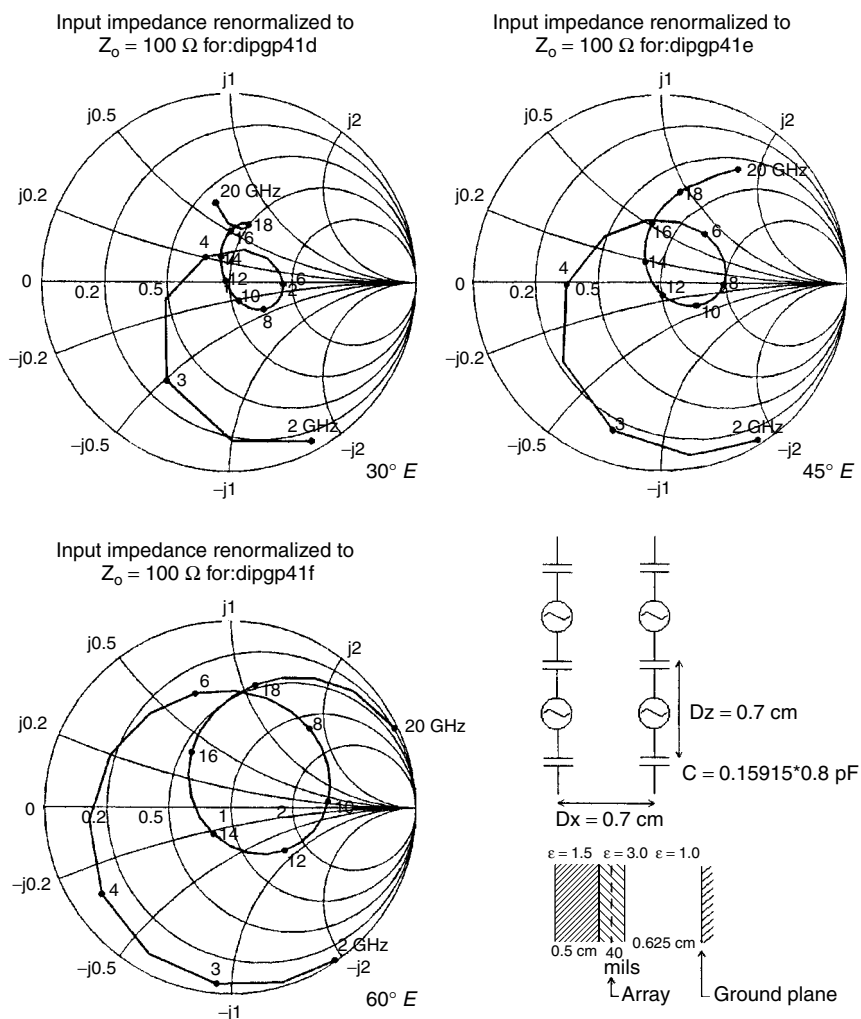


Fig. 6.9 Same array configuration as shown in Figs. 6.7 but for scan angles 30° , 45° , and 60° in the E plane.

and $\sim 0.09\lambda$ at 20 GHz), this section nevertheless plays an important role in obtaining the best match. It is simply a correction of the idealized situation shown in Fig. 6.6. It is also worth mentioning that the calculated curve is somewhat tighter than the qualitative one in Fig. 6.6. The reason for this is primarily that jX_A is somewhat more capacitive at the lower frequencies and more inductive at the higher; that is, jX_A is made somewhat more *narrowbanded* by itself in order to obtain greater bandwidth of the total scan impedance. This is counter to many people's intuition and is discussed further in Section 6.12.

Finally, we show in Fig. 6.8 the scan impedance for 30° , 45° , and 60° in the H plane. Similarly, Fig. 6.9 shows the scan impedances in the E plane for the same scan angles.

Note that the computed case has not been optimized for either bandwidth or scan compensation. This would be a good place to start. But please don't write a paper about it. It is simply left as an exercise for the students.

And by the way: Once you get this close to a good design, the winner among the various optimization programs is probably more a matter of coincidence than substance.

6.9 ARRAY WITH GROUNDPLANE AND TWO DIELECTRIC SLABS

In the previous section we demonstrated how an increase in bandwidth and/or reduction in VSWR could be obtained by placing a single dielectric slab in front of the array. It would be natural to expect further improvement if we used two or more slabs instead of a single slab. This is indeed the case. However, to obtain the potential improvement we shall need some design guidance, and for this purpose we will next show the transformations for one and two dielectric slab cases as seen in the rectangular coordinate system rather than the Smith chart. The reason for this choice is simply that it allows us to depict quite complicated transformations on a single piece of paper leading to a better overview. Thus, for the sake of comparison we show first in Fig. 6.10 the rectangular transformation of a single dielectric slab similar to the case shown earlier in the Smith chart in Fig. 6.6. We have chosen a dielectric constant for the slab equal to $\epsilon_1 = 2$, leading to a transformation of $2R_{A0}$ into R_{A0} at the center frequency f_0 . Thus, the transformation circle for all frequencies at broadside is given by the circle going through these two points as shown in Fig. 6.10. A typical input impedance of the transmission line looking left is as shown in the schematic denoted Z_{1-} . We next add the groundplane impedance Z_{1+} in parallel and obtain the point $Z_{1-} \parallel Z_{1+}$ located in the upper half of the complex plane. Finally, we add the antenna reactance jX_A and obtain the total scan impedance denoted $Z_{1-} \parallel Z_{1+} + jX_A$.

Next we show in Fig. 6.11 a typical two-layer case. Again, we have chosen the dielectric constants such that the impedance $2R_{A0}$ will be transformed into R_{A0} at the center frequency f_0 like in the single slab case. However, we must in the double-layer case choose still another parameter. We have chosen the impedance level between the two slabs to be midway between $2R_{A0}$ and R_{A0} ; that is, it is

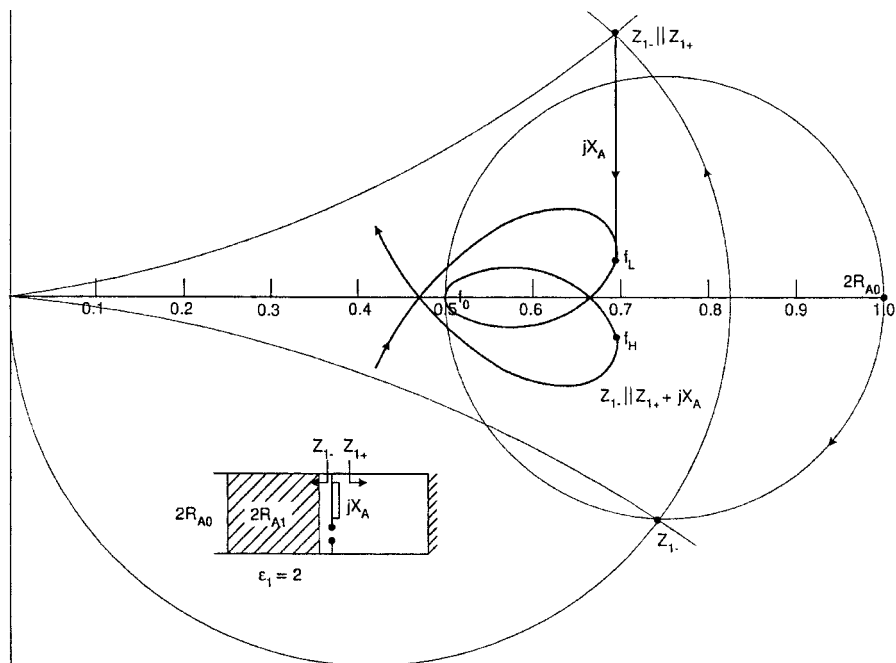


Fig. 6.10 Single dielectric slab. The input impedance as seen through the dielectric slab at the array in the negative direction is denoted Z_{1-} . As shown in the schematic, we then connect the groundplane impedance Z_{1+} in parallel and obtain $Z_{1-} \parallel Z_{1+}$. Finally, we add the antenna reactance jX_A in series and obtain the compensated terminal impedance curve $Z_{1-} \parallel Z_{1+} + jX_A$.

fixed at $1.5R_{A0}$. That completely determines the characteristic impedances of both dielectric slabs as indicated in Fig. 6.11 as well as the relative dielectric constants shown in the insert. More precisely, the transformation circles for the outer slab Z_1 will be determined as having a diameter going from $2R_{A0}$ to $1.5R_{A0}$, while the transformation circle for the second slab Z_2 will span over R_{A0} and $1.5R_{A0}$. The input impedance of the outer slab looking in the negative direction is denoted Z_{1-} . At a typical frequency f_L lower than f_0 it will be located on the lower half of the transformation circle for the first slab as indicated in the figure. We must next use this impedance Z_{1-} as the load impedance for the second slab. Since we also know the characteristic impedance Z_2 for the second slab, we can determine the transformation circle as shown in Appendix A, Case III (in particular, consult Fig. A.3). The pertinent construction lines are shown dashed in Fig. 6.11. Next, we determine the position of the input impedance Z_{2-} for the second slab as explained in Appendix A, Section A.2, Case III, in conjunction with Fig. A.3. Similar to the single-slab case we now add the groundplane impedance Z_{2+} in parallel and obtain in that way the point $Z_{2-} \parallel Z_{2+}$ in the upper part of the complex plane as shown. Finally, we add the antenna reactance jX_A in series and obtain the total scan impedance denoted as $Z_{2-} \parallel Z_{2+} + jX_A$.

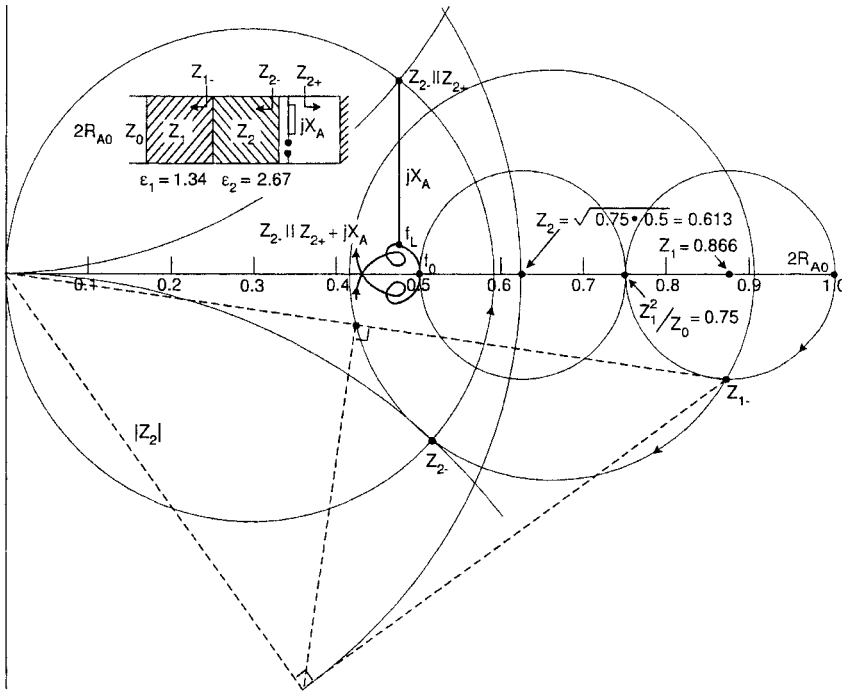


Fig. 6.11 Double dielectric slab. We first obtain the input impedance Z_{1-} of slab 1 looking left. Next, we find the input impedance Z_{2-} of the second slab when terminated in Z_{1-} . From the schematic we then see that the groundplane impedance Z_{2+} should be added in parallel yielding $Z_{2-} \parallel Z_{2+}$. Finally, we add the antenna reactance jX_A in series and obtain the compensated terminal impedance $Z_{2-} \parallel Z_{2+} + jX_A$.

6.10 COMPARISON BETWEEN THE SINGLE- AND DOUBLE-SLAB ARRAY

The determination of the locus for the scan impedance for a single slab in Fig. 6.10 and a double slab in Fig. 6.11 is only qualitative. However, the reduction of VSWR and to some extent an increase in bandwidth in the double-slab case is quite obvious. Inspection of the figures of the two cases shows that the primary reason is that the double-slab case produces a smaller variation of the real part of the scan impedance while the imaginary part must be balanced very carefully with the groundplane reactance Z_{2+} and the antenna reactance jX_A . In other words, a parametric study of actual calculated curves should be undertaken. A typical example will be given in the next section.

6.11 CALCULATED SCAN IMPEDANCE FOR ARRAY WITH GROUNDPLANE AND TWO DIELECTRIC SLABS

Undoubtedly, many readers have not felt too comfortable being “dragged around” the complex plane as was just demonstrated above. For those who prefer exact

Input impedance renormalized to $Z_0 = 100 \Omega$ for: dipgp68o

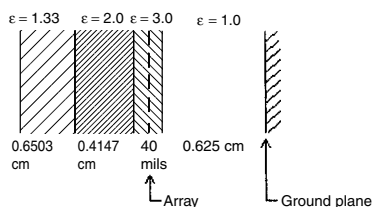
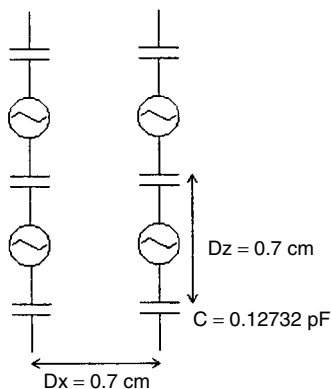
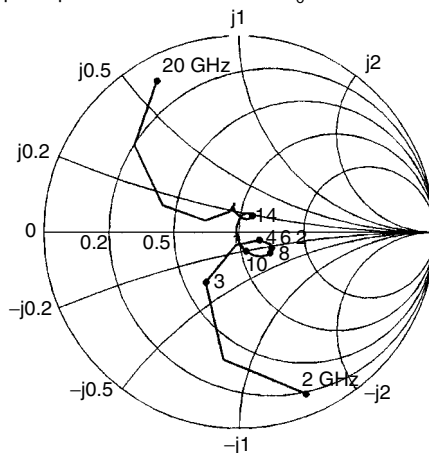


Fig. 6.12 Calculated impedance of a double-slab array as obtained from the PMM code with dimensions as shown in the insert. This curve also incorporates a small matching section comprised of a transmission line of length 0.12 cm and characteristic impedance 200 ohms. Note: This case has not been optimized.

calculated impedance curves, a typical calculated scan impedance curve is shown in Fig. 6.12. It is the result of an exploratory parametric study that showed that minimum VSWR is obtained by choosing the impedance level at the elements slightly higher than R_{A0} as was the case in Fig. 6.11. Rather, it should be chosen

around $1.32R_{A0}$. Similar to the single dielectric case in Fig. 6.7, it makes use of a short transmission line of length 0.12 cm and characteristic impedance 200 ohms in order to better center and compress the impedance curve.

As stated in the previous section, it clearly leads to a lower VSWR rather than increasing the bandwidth. However, it must be pointed out that no formal optimization for either one of these two features was performed.

Finally, we show in Fig. 6.13 the scan impedance for scanning at 30° , 45° , and 60° , respectively, in the H plane, while Fig. 6.14 shows the E -plane scan at the same scan angles. Compared to the single-slab cases shown in Figs. 6.8 and 6.9, we observe that the double-slab case continues to be superior to the single-slab

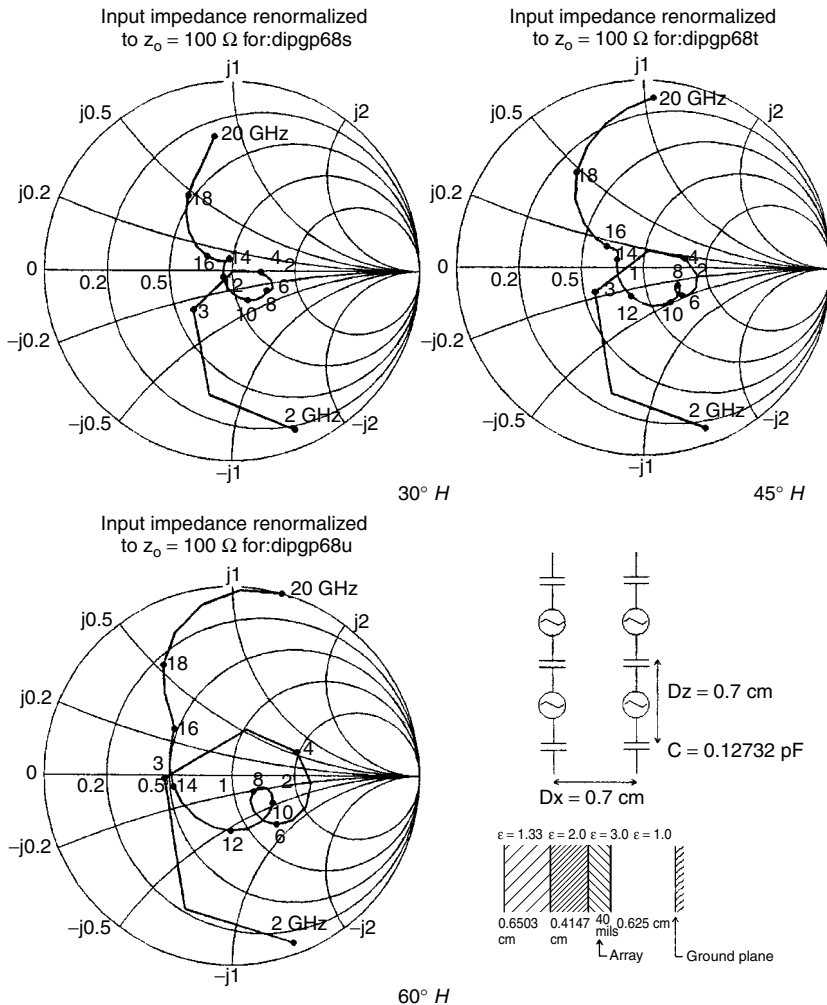


Fig. 6.13 The scan impedance for the same case as shown in Fig. 6.12 but for scan angles equal to 30° , 45° , and 60° in the H plane.

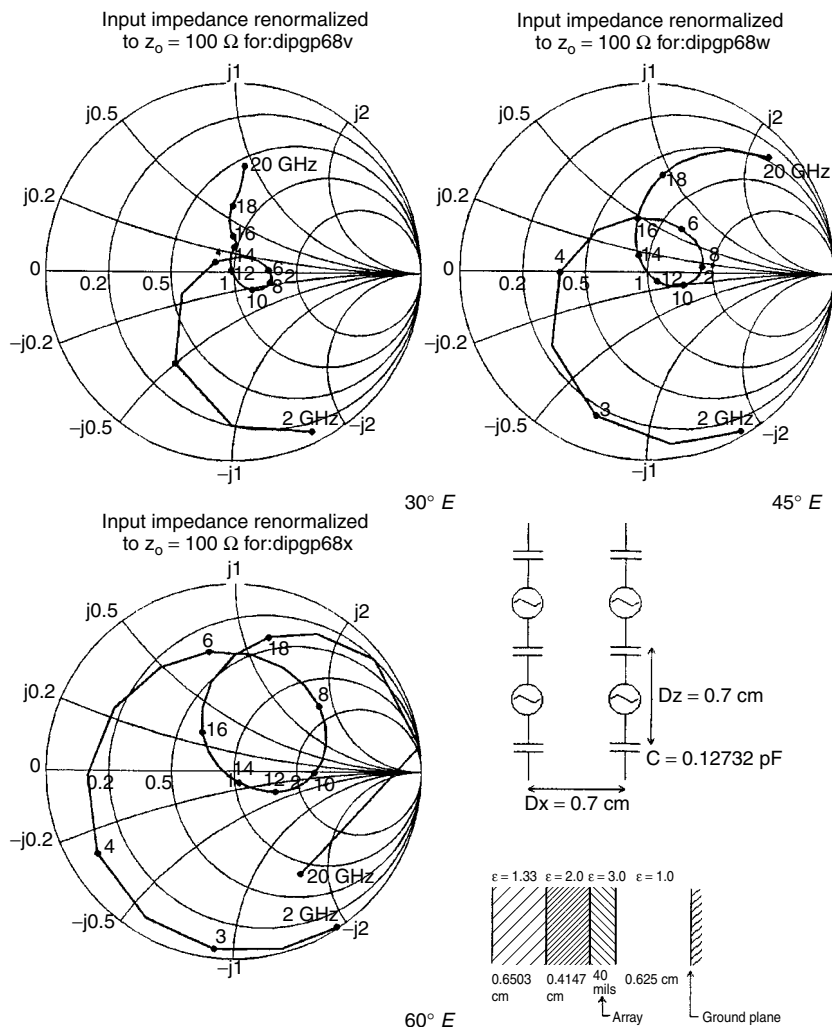


Fig. 6.14 The scan impedance for the same case as shown in Figs. 6.12 and 6.13 but for scan angles 30° , 45° , and 60° in the E plane.

case. However, we emphasize that no parametric study has been undertaken to determine a better design from a scan angle point of view. This could perhaps be worthy of a paper.

6.12 COMMON MISCONCEPTIONS

6.12.1 Design Philosophy

6.12.1.1 Array Versus Element It is often suggested that design of a broadband array starts with the development of a single element with large bandwidth.

Certainly, an element with an inherently small bandwidth is not very likely to yield a much wider bandwidth when placed in an array environment. The patch element would fall in that category. But a single short dipole is not particularly broadbanded either, although it is better than the typical patch (see also Appendix D). Still, when dipoles are placed in an array environment, a significant increase in bandwidth can be obtained as demonstrated by several examples earlier in this chapter. The trick consists of placing the dipoles close to each other as shown in Figs. 6.2 and 6.4 while the patches in general will be too large to do the same. To many designers, placing dipoles closer to each other runs counter to their intuition since it definitely increases the mutual coupling between elements. The physical explanation of this dilemma is simply to note that the stored energy associated with a single element removed from the array occupies the space reaching from the surface of the element to infinity. In contrast, the stored energy associated with a single element in an array reaches only from the element surface and midway to the next neighbor elements—that is, less space resulting in less stored energy. And that simply in general means more bandwidth (and for the record, this is also the reason “fat” dipoles have more bandwidth than the thin ones). See also Sections 6.3 and 6.12.2.

This argument does in no way contradict the fact that most stored energy is located in the immediate vicinity of the elements. Recall (see Sections 4.8–4.10 in reference 1) it is directly associated with the evanescent waves that decrease exponentially but are still “felt” at a typical distance of $\sim\lambda/4$ from the elements. Thus, if the interelement spacing typically is 0.3λ , the “storage space” is limited to 0.15λ .

[It is more complicated to examine the field from a single elements where the near field falls off like $1/r^2$ and $1/r^3$.]

Finally, the designer should be reminded that the element spacing should always be sufficiently small to avoid grating lobes and surface waves in dielectric slabs adjacent to the dipole elements.

6.12.1.2 Array Versus Groundplane Another misconception concerning a design approach is that one should first attempt to design a broadband array without a groundplane and then later battle the effect of adding a groundplane. Typically, that philosophy might lead to an array of closely wound flat spiral elements. Such an array without a groundplane can indeed have a reasonably low VSWR over a significant bandwidth as indicated in Fig. 6.15. However, note that the antenna impedance rotates several times around the Smith chart, in sharp contrast to the variation of a dipole array that goes slowly and monotonically from the capacitive to the inductive region (see Figs. 6.2 and 6.3). As we saw earlier in this chapter, this simply means that the antenna reactance jX_A and the groundplane impedance Z_{1+} could be designed to partly cancel each other and thereby yield a larger bandwidth as shown earlier. In contrast, the spiral array as well as other antennas, like the Vivaldi, can improve only when it happens to be in the capacitive region when Z_{1+} is inductive; otherwise the VSWR will get worse.

The fast variation with frequency as observed in the spiral array is typical for elements consisting of long wires (namely in this case the two arms of the

Long Wires vs. Short End-Loaded Dipoles

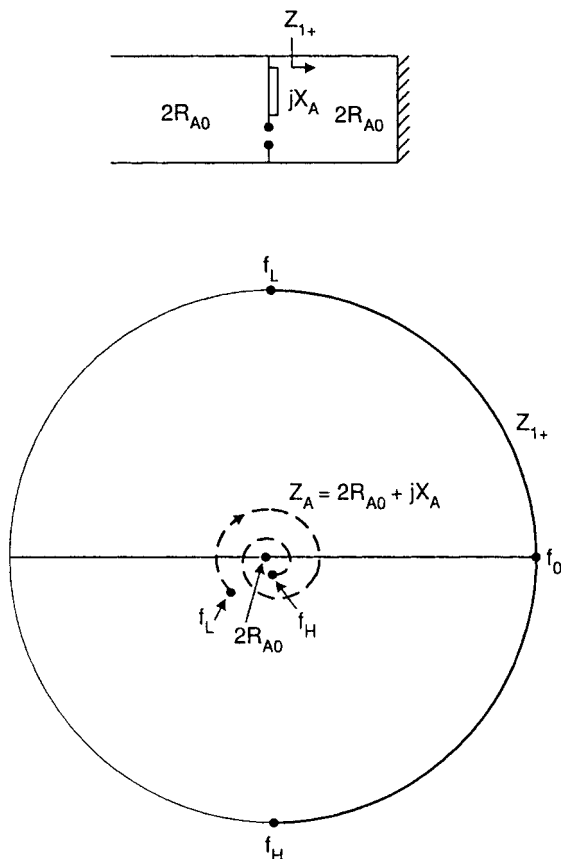


Fig. 6.15 Typical antenna impedance $Z_A = 2R_{A0} + jX_A$ in the negative direction for a long wire antenna like a flat spiral without the effect of a groundplane. At lower frequencies the groundplane impedance Z_{1+} is inductive as shown on the rim of the Smith chart. The total antenna may improve if jX_A is capacitive but get worse if inductive. Thus, no broadband compensation will take place.

spirals). It is quite similar to the input impedance of a long transmission line terminated in a load. However, in the spiral array case the loss should come primarily from radiation and not from actually terminating the arms in resistive loads (moderate loading might be acceptable to improve the performance of the spiral array at the lowest frequencies).

6.12.1.3 The “Moving Groundplane” (That Moves Too Fast!) We considered above an array of spirals in front of a groundplane. We observed how the antenna reactance X_A was incompatible with the groundplane impedance Z_{1+} and therefore would not lead to a consistently lower VSWR. However, one may

ask, Would it be possible to just make the groundplane impedance Z_{1+} very large over a large frequency range, if not infinite. In other words, could we somehow manage to keep the electrical spacing between groundplane and array equivalent to one quarter-wavelength at *all* frequencies?

One scheme often suggested to accommodate such an approach is shown in Fig. 6.16. We have placed an array of dipoles in front of a groundplane at a distance of one-quarter wavelength at the lowest operating frequency, for instance at $f_1 = 1$ GHz. In between the array and the groundplane is an FSS resonating at a higher frequency, such as $f_2 = 2$ GHz. The distance between the array and the FSS is one-quarter wavelength at f_2 as indicated in the figure.

We may further assume that the FSS is *transparent* at $f_1 = 1$ GHz (this is a gross approximation, but since this scheme is a scam anyway, so what!). Under that assumption we obtain $Z_{1+} = \infty$ at $f_1 = 1$ GHz as indicated in the Smith chart in Fig. 6.16. (A more rigorous calculation will show that the distance between the array and the groundplane should be somewhat less than $\lambda_1/4$ when we take into account the capacitive effect of the FSS at $f_1 = 1$ GHz.)

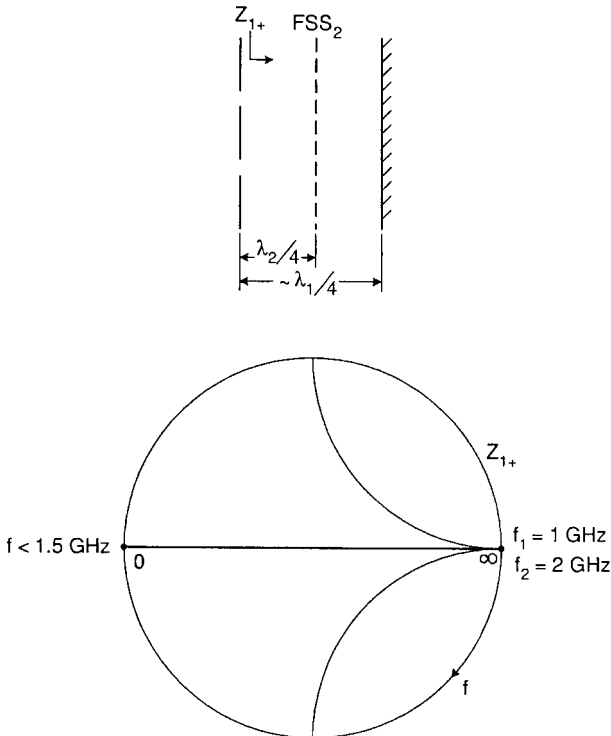


Fig. 6.16 The “moving” groundplane that moves too fast! An FSS resonating at 2 GHz will produce $Z_{1+} = \infty$ if the spacing between array and FSS is $\lambda_2/4$. Similarly, $Z_{1+} = \infty$ at $f = 1$ GHz, if the array and the groundplane is spaced slightly less than $\lambda_1/4$ from the array (see text). However, $Z_{1+} = 0$ somewhere between 1 and 2 GHz (Foster’s Reactance Theorem [101]).

Similarly, $Z_{1+} = \infty$ at $f_2 = 2$ GHz as also indicated in the same Smith chart. So far we have obtained our objective: Z_{1+} is infinite at 1 and 2 GHz. The question now is, What will happen at other frequencies? First of all we note that all components to the right of the array are lossless. Thus, Z_{1+} must be purely imaginary—that is, located somewhere on the rim of the Smith chart. It is now a well-established fact that all impedances at the *rim* of a Smith chart will move clockwise with increasing frequency (it cannot just stop!). Thus, when going from 1 to 2 GHz we simply must follow the rim of the Smith chart; that is, at some frequency between 1 and 2 GHz, Z_{1+} must pass through zero in the Smith chart, which of course is a clear disaster. (For those who need a more rigorous proof, consult *Foster's Reaction Theorem* [101].)

Thus, it may be safe to state that a groundplane that moves with frequency such that the spacing between the moving groundplane and the array is $\sim \lambda/4$ simply violates fundamental physical laws. It is not that the groundplane does not move; rather, it moves about twice as fast as we want! Then of course there are those who would turn that feature into a multiband affair!

6.12.1.4 A Little Loss Can Cure Anything! If one's main objective is to avoid going through zero in the Smith chart, it is certainly possible using the approach shown in Fig. 6.17. We have now added a load resistance R_L to the terminals of the FSS (or call it an array of loaded dipoles). This will result in an impedance of this sheet alone denoted by Z_{absorber} as indicated in the Smith chart in Fig. 6.17. If the spacing between the loaded FSS and the groundplane is $\lambda/4$ at the center frequency, the groundplane impedance Z_g as seen at the loaded array will look like also shown in the same Smith chart. Combining Z_{absorber} and Z_g in parallel yields the impedance Z_{2+} . Finally, if the spacing between Z_{absorber} and the actual array is $\lambda/4$ at the center frequency, the loaded groundplane impedance Z_{1+} is readily obtained by appropriate transmission line transformation of Z_{2+} in the Smith chart (see Appendix B).

Certainly, we have produced a loaded groundplane impedance Z_{1+} that does not go through zero over a considerable frequency range. However, we have in effect merely produced an absorber, or more precisely a circuit analog absorber. For an in-depth discussion about this type, see Chapter 9 in reference 102.

Typically, about half the input power will be absorbed and converted into heat (i.e., about a 3-dB loss). Thus, this is not a solution that is recommended in general—that is, unless you have a couple of hundred watts (or more) of transmitting power available and want to make an omelet (works best if you point the antenna toward nadir).

6.12.2 On the Controversy Concerning Short Dipoles

The self-impedance of a short dipole is indeed comprised of a low radiation resistance in series with a high capacitive reactance. Thus, when it is suggested to use these elements to build a broadband array, it is quite often discarded as being a bad idea.

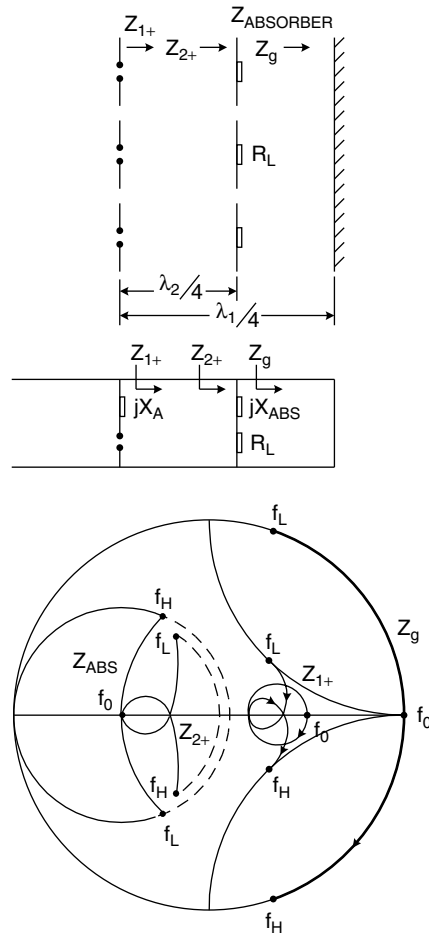


Fig. 6.17 By adding loss to the FSS in Fig. 6.16 either in the form of lumped resistors or otherwise, we can avoid going through (0,0). However, we have merely produced an absorber, resulting in losses that usually are unacceptable.

And when it subsequently is suggested to add an extra capacitance across the tips of the dipoles, many (or at least some) instantly assume that “Voodoo Physics” has made its entrance into the world of phased arrays.

Notwithstanding the fact that numerous examples to the contrary were given earlier in this chapter, there are nevertheless many readers that would appreciate a more direct physical discussion of this apparent dilemma.

Thus, to accommodate this segment of our readers, let us consider the development depicted in Fig. 6.18. First to the far left (Fig. 6.18a) we show an array comprised of $\lambda/2$ dipoles. It resonates at the frequency f_0 . The equivalent circuit of this array is shown in Fig. 6.18b. Note that it is comprised of equivalent inductances L_0 near the terminals and equivalent capacitances C_0 between the

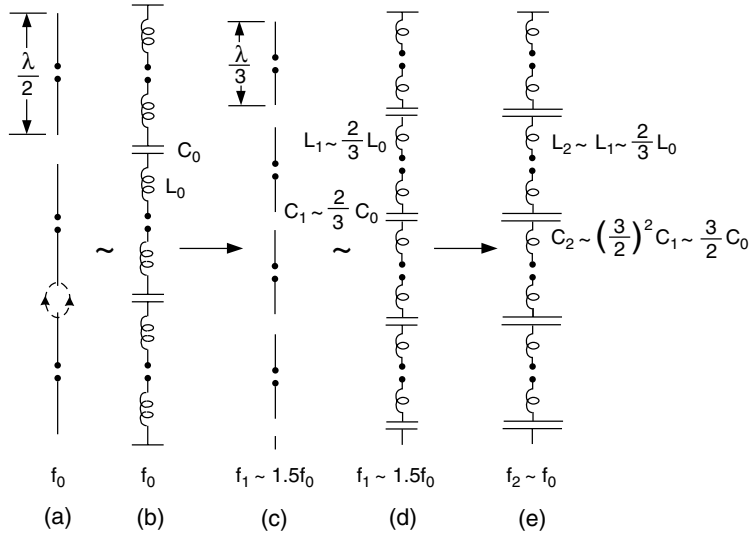


Fig. 6.18 Self-impedance of a short dipole is indeed capacitive. However, it contains no “big bad capacitance.” In fact, a dipole is comprised of an inductive wire L_0 near its terminals and a capacitance C_0 at the tips. Thus, if L_0 is reduced, C_0 must be increased to maintain resonance. Any conclusion to the contrary is based on voodoo physics. This figure explains why the capacitance between dipole tips must be increased if the dipoles are shortened and the resonance frequency is to be maintained.

dipole tips (that a short wire is essentially an inductance should be well known to even inexperienced antenna engineers, and the formula for tip capacitance is given by Weber [89]).

Next, in Fig. 6.18c we scale the entire array downward by the factor $\lambda/3$: $\lambda/2 \sim 0.67$; that is, this scaled array resonates at $f_1 = 1.5f_0$. In Fig. 6.18d we indicate that the new equivalent inductance is now $L_1 \sim 2/3L_0$ while the new equivalent capacitance is $C_1 \sim 2/3C_0$. Thus, based on these equivalent values, the array resonates at the correct frequency $f_1 = 1.5f_0$.

Finally, in Fig. 6.18e we show the same array as in Fig. 6.18d, but in Fig. 6.18e we have added an extra capacitance from tip to tip (i.e. in parallel with C_1) such that we obtain a new larger total capacitance equal to $C_2 \sim (3/2)^2 C_1 \sim 3/2 C_0$.

Thus, the resonant frequency for this new array with augmented tip capacitance is equal to f_0 .

Finally we remind the reader that the radiation resistance R_A of an array of dipoles without a groundplane is given by [103]

$$R_A = \frac{Z_0}{2D_x D_z} \frac{1}{r_y} [\perp P_{\perp} P_{\perp}^t + \parallel P_{\parallel} P_{\parallel}^t], \quad (6.1)$$

where D_x , D_z denotes the interelement spacings, r_y denotes the direction cosine of the beam direction with respect to the boresight axis, $\perp P_{\perp}$, $\parallel P_{\parallel}$ are the orthogonal

and parallel pattern components under scattering condition, respectively, and ${}_{\perp}P^t, {}_{\parallel}P^t$ are the orthogonal and parallel pattern components under transmitting conditions, respectively.

When an array is scaled, everything in (6.1) remains essentially constant except of course the interelement spacings D_x and D_z .

Thus, if D_x and D_z are reduced, the radiation resistance R_A will increase; in fact for close spacings it may easily reach several hundred ohms.

The fallacy with the short dipole argument alluded to above is simply that many readers do not fully appreciate how dominating a role the mutual coupling can play. Historically, it is rooted in the fact that the classical phased array approach typically would start by considering only a single element of whatever type. It was then hoped that the mutual coupling would be so low that the features of a single element would not be “masked” too badly in the array environment. Otherwise it became necessary to calculate the entire array which was in general frowned upon.

Today we know that it is actually easier to calculate the terminal impedance of an infinite array than it is to calculate it for a single element. And the mutual coupling should not be viewed as an evil to be avoided. On the contrary, when handled properly, it can be a true blessing to be enjoyed. And you certainly cannot outlaw it.

P.S.: The mutual coupling between adjacent dipoles in the typical arrays considered here is very large. We do not know how large because we do not care! And there is not much you can do with it anyway except marvel at it.

6.12.3 Avoid Complexities

6.12.3.1 On Array Gain Some readers may by now have noticed that very little attention has been given to the gain of the array. This should not be interpreted as lack of interest in that subject. Rather it reflects the fact that it can be obtained in a forward and simple manner.

It was already pointed out in Chapter 2, Section 2.6.2, that a large array with groundplane, uniform illumination, and physical area A also has a receiving area equal to A . Thus, if the array is designed to have no grating lobes and no significant surface waves, we may conclude from one of the most fundamental laws in antenna theory that the directivity D of the array is

$$D = \frac{4\pi}{\lambda^2} A. \quad (6.2)$$

If the aperture illumination is tapered, the directivity in (6.2) should be multiplied by the aperture efficiency. That depends only on the aperture distribution [104].

Finally, the gain G is obtained from (6.2) by multiplication of the loss efficiency. Most commonly we are here talking about transmission line loss in the feeding harness and possibly some from the dielectric substrate supporting the elements.

Also, we must add the mismatch loss. If the antenna has a $VSWR < 2$ (as typical for the arrays discussed here), the mismatch loss will be less than 0.5 dB.

We emphasize that our derivation fails if there are grating lobes or a strong presence of surface waves. Both of these calamities are usually recognized by their adverse impedance behavior. Again we remind the reader that the most dangerous surface waves are the ones that show up only in the finite arrays (see Chapters 4 and 5). However, as also explained there, these surface waves can be greatly attenuated by feeding the elements from voltage generators with realistic generator impedances.

Thus, it is possible to get a relatively quick handle on the gain of a large and well-designed array. However, not all designers see it that way. Confusion often occurs if one tries to obtain the “gain” of a single element in the array. This can be made into a highly complex endeavor and will therefore often be the source of controversial results. The author remembers in particular a case where it was claimed that the gain of an array was low because the “element currents were low.”

This is quite a common misconception. The amplitude of the element currents has nothing to do with the gain. Usually it just means that the element impedances are high. However, if the array impedance is properly matched, the field strength observed in the far field will be determined entirely by the total gain of the array that is dominated by the array factor and not the element pattern as discussed in *Common Misconceptions*, Section 2.14.3. See also Section 6.12.3.2.

For those who still prefer to marvel over the element pattern and gain, just take the total array gain as calculated above and divide it by the number of elements. At some point down the road, they will most likely ask for the total array gain. Well, just multiply the element gain by the number of elements!

6.12.3.2 Will More Elements Increase the Gain? It is sometimes implied that adding more elements to a large array of a given aperture will increase the gain. Well, under some circumstances it will and under others it won't. Let us examine this phenomena in a little more detail.

Consider first an array where the density of the elements is so low that grating lobes can exist. Since energy is lost in the grating lobe(s) (obviously we are transmitting), we will have less power going into the mainbeam and consequently a lower gain in that direction. Thus, if grating lobes can exist, adding more elements will increase the gain (the author can think of only a few cases where the presence of a grating lobe has been any blessing to mankind).

Let us now look at the case where the element density is so large that no grating lobe can exist. In that case the radiation resistance R_A of each element is given by [103]

$$R_A = \frac{Z_0}{2D_x D_z} \frac{1}{r_y} P^t P, \quad (6.3)$$

where D_x and D_z are the interelement spacings in the X and Z directions, respectively, r_y is the direction cosine along the y axis of the main beam, and P^t and

P are the radiation patterns under transmitting and receiving conditions, respectively.

Let us further denote the number of elements per meter along the X and Z directions by

$$N_x = \frac{1}{D_x} \quad \text{and} \quad N_z = \frac{1}{D_z}, \quad (6.4)$$

respectively.

From (6.3) and (6.4) we then obtain the following equation for the energy being supplied to the array per square meter:

$$P_A = N_x N_z R_A I_A^2 = (N_x N_z)^2 \frac{Z_0 P^t P}{2r_y} I_A^2, \quad (6.5)$$

where I_A denotes the effective element currents.

The far field E_{far} in the mainbeam must be proportional to I_A , N_x and N_z , that is,

$$E_{far} = C N_x N_z I_A, \quad (6.6)$$

where C is merely a proportionality factor. Substituting in (6.6) for I_A obtained from (6.5), we obtain

$$E_{far} = C \sqrt{\frac{2 P_A r_y}{Z_0 P^t P}}. \quad (6.7)$$

Thus, for no grating lobes and no surface waves we observe that E_{far} is completely independent of the total number $N_x N_z$ of elements. Obviously so is the gain. That should come as no surprise.

Remember that the antenna pattern is dominated completely by the array factor (see Section 2.14.3). The elements are merely there to approximate a continuous aperture. Once they reach a certain density, we are in a sense just oversampling.

But isn't anything happening when we increase the number of elements? Certainly, namely the radiation resistance R_A increases [see (6.3)] while the element currents I_A are reduced [see (6.5)]. And most importantly, the stored energy is being reduced, which means more bandwidth.

Of course, this is what this chapter is all about.

It is sometimes implied that a weak element current I_A results in a low far field and thus a low gain. It is hoped that this section will put this notion to sleep. See also Section 6.12.3.1. And by the way, adding more elements to produce endfire elements does not help either!

6.12.4 What Is So Special About $\lambda/4$ Anyway?

This author has been known to state that in microwave technology good things happen if you move in one-quarter time (unless you are in Vienna of course, where it should be in three-quarter time).

More specifically, is there something special going on when the distance between an array and a groundplane is one-quarter wavelength? Well, everyone can quickly convince themselves that with a space delay of $2 \times 90^\circ$ and a 180° phase reversal in the perfectly electric conducting groundplane, the direct and the reflected signal will be in perfect phase in the forward direction. Apparently, many will assume that this is where the antenna gain is maximum. However, this is just another common misconception. To clarify this, consider the radiation pattern for a single column of dipoles in front of a PEC groundplane as shown in Fig. 6.19 at different frequencies. First of all we note that since the image in the groundplane is always 180° out of phase with the original, we will always observe a null along the groundplane. Furthermore, at the lowest frequency where the spacing to the groundplane is 0.06λ , the radiation pattern for the dipole column and the groundplane will be almost a perfect circle as shown. At the higher frequency where the groundplane spacing is 0.25λ , the radiation pattern deviates slightly

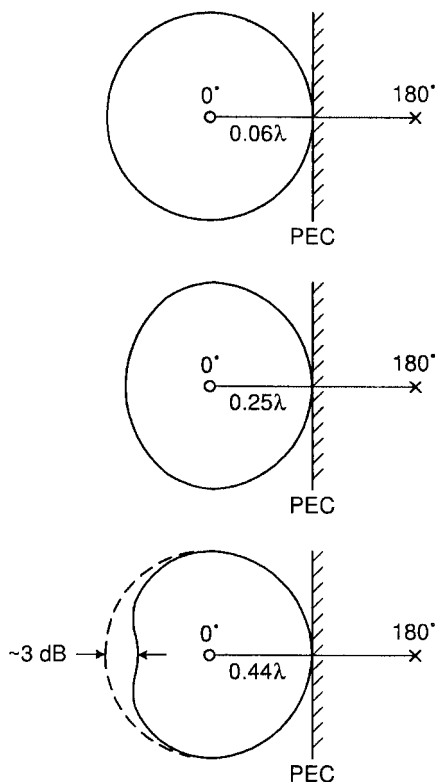


Fig. 6.19 Radiation pattern for a column array in front of a PEC groundplane seen end-on. By pattern integration we easily see that the directivity is largest at the lowest frequencies where the groundplane spacing is very small (but not too close to zero). At higher frequencies we obtain lower directivity. Note in particular that nothing special happened when the spacing is equal to $\lambda/4$.

from a circle, being a little broader. And when the spacing finally is $\sim 0.44\lambda$, the radiation pattern is about 3 dB lower in the forward direction than the maximum value. (To be sure, for a groundplane spacing of 0.5λ we would observe a null in the forward direction.) The directivities of those three patterns can be obtained by simple pattern integration. Without actually doing that, any seasoned antenna engineer would agree that the difference between these three cases would be a couple of decibels or so—that is, about the same over a frequency range of $0.44/0.06 = 7.3:1$. While it would be unwise to go up in frequency, no great problem would be encountered at a groundplane spacing of, for example, 0.04λ ; that is, we would then cover a frequency range of $0.44/0.04 = 11.0:1$. If we were to go further down in frequency, the radiation resistance would eventually get so low that ohmic losses could diminish the efficiency. However, noticing that the surface resistance for a good conductor (copper) is only 0.024 ohm per square at 10 GHz, that is usually not as significant a problem as many expect. (While working as an antenna engineer in industry in the early 1960, the author designed and built a communication antenna for the Titan missile system. The distance between the vertical dipoles and a very heavy mast was only $\sim 0.01\lambda$ at the lowest frequency.¹ It was with some anxiety that he watched the customer fly in a 1-kW transmitter and meticulously measure the temperature rise at seven agreed-upon checkpoints on the dipoles and the mast. While he does not remember the exact temperature rise, it simply turned out to be a non-event, much to everyone's surprise.) For further comments consult Kraus and Marhefka [104].

Finally, we should point out that when the individual elements are arranged into an array, the array pattern is so domineering compared to the element pattern that the directivity of the individual elements becomes almost of pure academic interest only. So what can we conclude? Simply that if you can match your antenna and stay away from the ohmic losses, you are probably OK! See also Sections 2.14.3 and 6.12.3.1.

P.S.: Actually, it is (or should be) well known that maximum directivity and gain are not necessarily obtained for the in-phase condition. For example, an endfire array of dipoles attains maximum endfire gain when the phases of the elements are delayed $\sim 3/N$ radians per element in addition to the simple end-fire condition. This is called the Hansen–Woodyard condition [105]. The W8JK antenna by Kraus is another example [106].

6.12.5 Would a Magnetic Groundplane Be Preferable to an Electric One (If It Were Available)?

The notion discussed above—namely, that *good things will happen* when the groundplane spacing is $\sim \lambda/4$ —sometimes erupts into other ingenious ideas. What about if we instead of a PEC groundplane would use one made of a perfect *magnetic* conductor (PMC)? In that case the image would be in phase with the

¹ The antenna had to fit into a silo during storage.

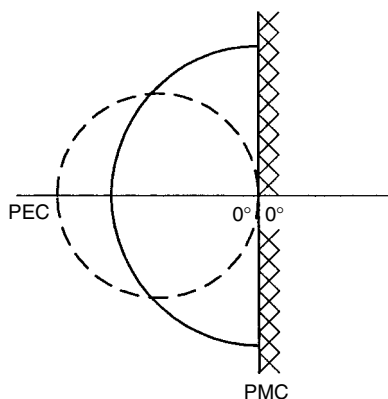


Fig. 6.20 Radiation pattern for a column array seen from the top mounted directly in front of a perfectly magnetic conducting (PMC) groundplane and, for comparison, the same if using a PEC groundplane at some distance (see Fig. 6.19). By pattern integration we observe that the former has lower directivity than the latter.

original if the latter were placed directly on the PMC as shown in Fig. 6.20. Consequently, the potential bandwidth could reach from DC to broad daylight!

However, before being carried away, let us consider the pattern of an electric dipole mounted in front of a PMC groundplane. Since the image always is in phase with the original, the total pattern would be the same as that of a single dipole—that is, a circle without a null as shown in Fig. 6.20. Also shown for comparison is the typical antenna pattern for the PEC groundplane considered above. Pattern integration will readily reveal that the *ideal* magnetic groundplane actually leads to lower directivity than the PEC counterpart.

But for those who still think that a Nobel prize is just around the corner, I must report about my most recent visit to RadioShack. Although not staffed with EE candidates, they nevertheless employ people with reasonably good technical knowledge and they do fairly well what they are supposed to do, namely, “move the merchandise.” One of them, however, was a little bit cocky and in one of my mischievous moments I decided to have a little fun. “Could I also have a roll of magnetic conductor?” I inquired. “Certainly,” came the quick answer as the young man disappeared into a back room. After a few minutes he reappeared and with great regret confided, “Sorry sir, we just ran out of it.” Too bad. I would have taken a couple of rolls.

But could we somehow by artificial means produce an equivalent magnetic groundplane? There has certainly been several attempts to do just that. However, everything this author has seen so far has in effect been merely an electric groundplane where the distance between the array and groundplane has been shortened by a more or less elaborate use of capacitive loading. This technique is actually well known to seasoned antenna engineers. So is the price: You pay in the form of a reduced bandwidth. You are in general much better off just staying with a plain electric groundplane as discussed earlier in this chapter.

6.12.6 Will the Bandwidth Increase or Decrease When a Groundplane Is Added to an Array?

It is quite common to hear statements to the effect that when a groundplane is added to an array the bandwidth is reduced. Apart from the fact that we have seen several examples earlier in this chapter where just the opposite was the case, it is interesting to shortly explore the reasons for this conclusion.

The most common argument is simply that when the distance between array and groundplane is one-quarter electrical wavelength, the signals from the array and the groundplane will be in phase in the forward direction (see also Section 6.12.4). Thus, it is concluded that this must constitute the *optimum*! And when the frequency deviates, it goes *downhill*; that is we will lose bandwidth.

This conclusion is wrong on several accounts. First of all, as discussed earlier in this chapter, there is no particular spacing that constitutes an optimum condition but rather a very broad range of frequencies where the directivity is almost independent of the distance to the groundplane as long as the distance does not get too close to one-half wavelength or zero. Second, since the groundplane impedance Z_{1+} is inductive at the lower frequencies while the array reactance X_A (for dipoles) is capacitive, the two components can as demonstrated earlier in this chapter by numerous examples be designed to partly cancel each other and thereby yield a greater bandwidth.

But what if we considered an array of slots rather than dipoles? Well, that would result in several interesting differences. First of all there would be only the signal coming directly from the slots, none would be coming directly from the groundplane (see Chapter 7 in reference 107). Thus, there would be no *optimum* groundplane spacing as far as radiation is concerned, although we could of course not make it equal to one-half wavelength or zero for matching reasons. However, as far as bandwidth is concerned, the most noteworthy would be the fact that the impedance of a slot array² is inductive at the lower frequencies and capacitive at the higher, which is, as we saw earlier, precisely the same way the groundplane impedance Z_{1+} behaves with frequency. Thus, the two impedance components will no longer cancel each other as in the dipole case but rather reinforce each other, leading to a narrower bandwidth when adding a groundplane to a slot array.

Interesting to note how it often is possible to arrive at the right conclusion by using the wrong argument! Or we may use intuition, which is great as long as it has been honed by experience and knowledge that conform to reality.

6.13 CONCLUSIONS

We have considered arrays of short dipoles with capacitors between their tips. By making the interelement spacing D_x small (typically 0.7 cm for the frequency range 2–18 GHz) it was possible to obtain an array impedance with basically just

² We assume that the total length of the slots is $\sim \lambda/2$.

one resonant frequency, a fairly constant real part R_A , and an imaginary part jX_A being capacitive below resonance and inductive above. Adding a groundplane was equivalent to adding a pure reactance Z_{1+} in parallel with jX_A . Since this reactance was inductive at the lower frequencies and capacitive at the higher, it could partly compensate for the frequency variations of jX_A and Z_{1+} leading to an increase of the bandwidth.

Further improvement in bandwidth could be obtained by placing a dielectric slab in front of the dipole array. When it was one-quarter wavelength thick, it acted merely as a transformer. However, at lower frequencies we experienced in addition to some resistive transformation also a reactive capacitive component. Thus, the reactance associated with the slab would be pulling in the same direction as the array reactance X_A ; that is, they would both pull in the opposite direction of the groundplane impedance Z_{1+} . Since the latter is the most domineering in the frequency range of interest here, it is possible to obtain just the right balance to achieve a very large bandwidth ($>7.0:1$).

Use of two or more dielectric slabs, each of which was approximately one-quarter wavelength thick (electrically), at the center frequency could lead to further improvement in bandwidth or lowering of the VSWR. However, a systematic optimization of the dielectric constants of the slabs and of the antenna as a whole was not done.

When this concept is presented for the first time, many would expect this design to be distinctly narrowbanded. It is usually based on the performance of a single, very short dipole that is known to be highly capacitive and very narrowbanded when tuned to resonance with an inductance. However, the most noteworthy point here is that it is the performance in the array environment that is important and not the individual elements. And due to a very strong mutual coupling between the elements, the array performance may differ vastly from the individual elements. In other words, while mutual coupling between elements historically has been treated as an evil, it can be a blessing when used correctly (see also Appendix D and Section 6.12.2).

Similarly, when thicknesses or spacings are prescribed to be one-quarter-wavelength at the center frequency, the performance is often expected to be mediocre at frequencies below and above the center frequency. While this often may be the case, we have demonstrated that it is indeed possible to play the various components out against each other and thereby obtain greater bandwidth.

Pursuit of this analytic design approach is in this writer's opinion where antenna engineers might demonstrate their foremost skill. It is not an art and is not necessarily based on years of experience. To the contrary, it is based on sound and fundamental engineering principles. Unfortunately, much design work today is turned over to a computer and eventually assisted by an optimization program, but is otherwise completely untouched by the human brain. While that approach can lead to some success, there is plenty of evidence to suggest that the truly optimum solution might be missed. The problem is that many designers today really do not have much of a clue of what should be optimized. A meaningful

optimization should merely be checking out minor tedious perturbations after you have a concept and a fundamental design. And that is precisely what it originally was intended to do.

It should finally be mentioned that the concept for this antenna as well as certain practical details are covered by US patent no. 6512487.